Inverse Laplace transforms

Normally when we calculate a Laplace transform, we start with a function $f(t)$ and we transform it into $F(s)$. Or in different variables, start with a function $y(x)$ and transform it into $Y(s)$. For instance, from the Laplace transform table, we know that e^t transforms to

$$\mathscr{L}(e^t) = \frac{1}{s-1}$$

Inverse transforms

Not surprisingly, an **inverse Laplace transform** is the opposite process, in which we start with $F(s)$ and transform it back to $f(t)$, or start with $Y(s)$ and transform it back to $y(x)$. So in this example, if we have

$$F(s) = \frac{1}{s-1}$$

we know that the inverse transform is $f(t) = e^t$. So if before we'd write the Laplace transform of $f(t)$ as $\mathscr{L}(f(t)) = F(s)$, we can now say that the inverse transform of $F(s)$ is

$$\mathscr{L}^{-1}(F(s)) = f(t)$$

The $\mathscr{L}^{-1}$ notation is what indicates the inverse Laplace transform. And as we saw with regular Laplace transforms, the inverse transform of a sum is the sum of the inverse transforms, and constant coefficients don't affect the value of an inverse transform.

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$$\mathscr{L}^{-1}(aF(s)+bG(s)) = a\mathscr{L}^{-1}(F(s)) + b\mathscr{L}^{-1}(G(s))$$

Ideally we want to simplify $F(s)$ to the point where we can compare each part of it to a formula from the table of Laplace transforms. If we can identify formulas in the table that match each part of $F(s)$, then we can transform back from s to t.

Very often though, we'll need to manipulate $F(s)$ first, because it won't be perfectly ready for an inverse transform. Let's do an example where we undo the transform, putting $F(s)$ back in terms of t.

Example

Use an inverse Laplace transform to find $f(t)$.

$$F(s) = \frac{s+7}{s^2-3s-10}$$

Our goal is to simplify $F(s)$ so that we can use formulas from the table of Laplace transforms. Currently, $F(s)$ is too complicated to match to any formula from the table, so we'll try to simplify it by factoring the denominator,

$$F(s) = \frac{s+7}{(s+2)(s-5)}$$

and then using a partial fractions decomposition.

$$\frac{s+7}{(s+2)(s-5)} = \frac{A}{s+2} + \frac{B}{s-5}$$

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To solve for A, we'll remove the $(s+2)$ factor, set $s=-2$, and find the value of the left side.

$$\frac{s+7}{s-5}$$

$$\frac{-2+7}{-2-5}=\frac{5}{-7}=-\frac{5}{7}$$

To solve for B, we'll remove the $(s-5)$ factor, set $s=5$, and find the value of the left side.

$$\frac{s+7}{s+2}$$

$$\frac{5+7}{5+2}=\frac{12}{7}$$

Plugging the values we found for A and B back into the partial fractions decomposition gives

$$F(s)=\frac{-\frac{5}{7}}{s+2}+\frac{\frac{12}{7}}{s-5}$$

Factoring the constants out of each numerator, we can rewrite the transform as

$$F(s)=-\frac{5}{7}\left(\frac{1}{s+2}\right)+\frac{12}{7}\left(\frac{1}{s-5}\right)$$

In this form, we can see that both values inside the parentheses resemble the Laplace transform

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$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

We'll just change the $1/(s+2)$ so that it matches the transform formula exactly.

$$F(s) = -\frac{5}{7}\left(\frac{1}{s-(-2)}\right) + \frac{12}{7}\left(\frac{1}{s-5}\right)$$

Now we can see that $a_1 = -2$ and $a_2 = 5$. We'll reverse the formula from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = -\frac{5}{7}(e^{-2t}) + \frac{12}{7}(e^{5t})$$

$$f(t) = \mathscr{L}^{-1}(F(s)) = \frac{12}{7}e^{5t} - \frac{5}{7}e^{-2t}$$

Because we used an inverse transform to work backward from $F(s)$ to $f(t)$, we know that if we'd started with this $f(t)$ and applied a Laplace transform to it, $\mathscr{L}(f(t))$, we would have found the transform $F(s)$ that we started with.

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